

① If $y = \sin^{-1}x$ then prove that

$$(1-x^2) Y_{n+2} - (2n+1)x Y_{n+1} - n^2 Y_n = 0$$

2) If $y = a \cos(\log x) + b \sin(\log x)$ then prove that

$$x^2 Y_{n+2} + (2n+1)x Y_{n+1} + (n^2+1) Y_n = 0$$

3) Obtain a reduction formula for

$$\int \sin^n x \, dx \text{ and hence evaluate } \int \sin^6 x \, dx$$

4) Obtain a reduction formula for $\int \sec^n x \, dx$ and hence evaluate

$$\int \sec^7 x \, dx$$

5) Using Beta, Gamma function evaluate

$$\textcircled{1} \int_0^a x^2 (a^2 - x^2)^{3/2} dx$$

$$\textcircled{2} \int_0^{\pi/2} \sin^4 x \cos^2 x \, dx$$

⑧ Find the 1st and 2nd partial derivative of

$$f(x, y) = x^2 - y^2 + 3xy$$

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evaluate ① $\int_0^a x^2 (a^2 - x^2)^{3/2} \, dx$

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Q) show

Q) Evaluate $\int_1^2 \int_0^x \frac{dy dx}{y^2 + x^2}$

Q) Evaluate $\int_1^8 \int_0^{\log y} x dx dy$

Q) Verify Euler's theorem for the function

① $u = f(x, y) = ax^2 + 2hxy + by^2$

② $u = x^n \sin\left(\frac{y}{x}\right)$

Q) If $u = \log(x^3 + y^3 + z^3 - 3xyz)$ show that

① $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \frac{3}{x+y+z}$

② $\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right) u = \frac{9}{(x+y+z)^2}$